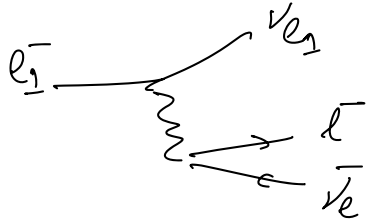


lepton flavor universality of weak interactions

$$g_w^e \equiv g_w^\mu \equiv g_w^\tau$$

$$\tau^- \rightarrow \mu^- \bar{\nu}_\mu \nu_\tau / e^- \bar{\nu}_e \nu_\tau$$

$$\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$$



$$l_1 = \tau^-, \mu^-$$

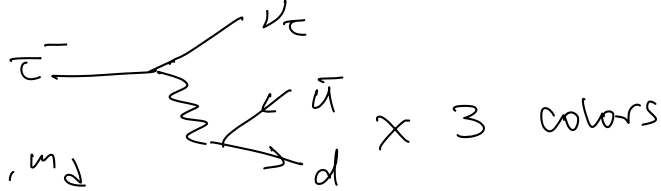
$$m_\tau > m_\mu > m_e$$

$$l_2 = \mu^-, e^-$$

$$\Gamma_{\text{tot}}^\tau = \Gamma_{\tau \rightarrow \mu} + \Gamma_{\tau \rightarrow e} + \Gamma_{\tau \rightarrow \text{had}} \times 3 (3 = N_C)$$

$$m_\tau = 1.78 \text{ GeV}$$

$$m_\tau \gg m_\mu, m_e, m_u, m_d$$



$$\Gamma_{\tau \rightarrow e} \simeq \Gamma_{\tau \rightarrow \mu} \simeq \Gamma_{\tau \rightarrow \text{had}}$$

$$\Gamma_{\text{tot}}^\tau \simeq 5 \Gamma_{\tau \rightarrow e}$$

what is the lifetime of τ ?

$$\tau_\mu = 2.2 \times 10^{-6} \text{ s}$$

τ^- DECAY MODES	Fraction (Γ_i/Γ)	Scale factor/ Confidence level	p (MeV/c)
Modes with one charged particle			
particle $^- \geq 0$ neutrals $\geq 0 K^0 \nu_\tau$ ("1-prong")	(85.24 $\pm$ 0.06) %		—
particle $^- \geq 0$ neutrals $\geq 0 K_L^0 \nu_\tau$	(84.58 $\pm$ 0.06) %		—
$\mu^- \bar{\nu}_\mu \nu_\tau$	[g] (17.39 $\pm$ 0.04) %		885
$\mu^- \bar{\nu}_\mu \nu_\tau \gamma$	[e] (3.67 $\pm$ 0.08) $\times 10^{-3}$		885
$e^- \bar{\nu}_e \nu_\tau$	[g] (17.82 $\pm$ 0.04) %		888
$e^- \bar{\nu}_e \nu_\tau \gamma$	[e] (1.83 $\pm$ 0.05) %		888
$h^- \geq 0 K_L^0 \nu_\tau$	(12.03 $\pm$ 0.05) %		883
$h^- \nu_\tau$	(11.51 $\pm$ 0.05) %		883
$\pi^- \nu_\tau$	[g] (10.82 $\pm$ 0.05) %		883
$K^- \nu_\tau$	[g] (6.96 $\pm$ 0.10) $\times 10^{-3}$		820
$h^- \geq 1$ neutrals ν_τ	(37.01 $\pm$ 0.09) %		—
$h^- \geq 1 \pi^0 \nu_\tau$ (ex. K^0)	(36.51 $\pm$ 0.09) %		—
$h^- \pi^0 \nu_\tau$	(25.93 $\pm$ 0.09) %		878
$\pi^- \pi^0 \nu_\tau$	[g] (25.49 $\pm$ 0.09) %		878
$\pi^- \pi^0$ non- $\rho(770) \nu_\tau$	(3.0 $\pm$ 3.2) $\times 10^{-3}$		878
$K^- \pi^0 \nu_\tau$	[g] (4.33 $\pm$ 0.15) $\times 10^{-3}$		814
$h^- \geq 2 \pi^0 \nu_\tau$	(10.81 $\pm$ 0.09) %		—
$h^- 2 \pi^0 \nu_\tau$	(9.48 $\pm$ 0.10) %		862
$h^- 2 \pi^0 \nu_\tau$ (ex. K^0)	(9.32 $\pm$ 0.10) %		862
$\pi^- 2 \pi^0 \nu_\tau$ (ex. K^0)	[g] (9.26 $\pm$ 0.10) %		862
$\pi^- 2 \pi^0 \nu_\tau$ (ex. K^0),	< 9 $\times 10^{-3}$ CL=95%		862

$$\frac{\Gamma(\tau \rightarrow e)}{\Gamma(\mu \rightarrow e)} = \frac{m_\tau^5}{m_\mu^5} \frac{G_F^2}{G_F^2} \frac{\rho(\tau \rightarrow e)}{\rho(\mu \rightarrow e)}$$

$$m_\tau \gg m_e, m_\mu \gg m_e \Rightarrow$$

$$\Gamma(\tau \rightarrow e) = \Gamma(\mu \rightarrow e) \frac{m_\tau^5}{m_\mu^5}$$

$$B(\mu \rightarrow e) \simeq 100\% \Rightarrow \Gamma(\mu \rightarrow e) \simeq \Gamma_{\text{tot}}^\mu = \frac{1}{\tau_\mu}$$

$$\Gamma(\tau \rightarrow e) \simeq \frac{1}{5} \Gamma_{\text{tot}}^\tau = \frac{1}{5} \frac{1}{\tau_\tau}$$

Experimentally we can measure lifetimes.

$$\frac{1}{5} \frac{1}{\tau_\tau} = \frac{1}{\tau_\mu} \left(\frac{m_\tau}{m_\mu} \right)^5 \Rightarrow \tau_\tau^{\text{pred.}} = \frac{1}{5} \tau_\mu \cdot x \left(\frac{m_\mu}{m_\tau} \right)^5$$

$$\frac{m_\mu}{m_\tau} = \frac{106 \text{ MeV}}{1780 \text{ MeV}}$$

$$\Rightarrow \tau_\tau^{\text{pred.}} \approx 3.3 \times 10^{-13} \text{ sec} \quad \text{prediction with weak int. ass. lepton flav. univ.}$$

$$\tau_{\text{PDG}} = 3.1 \times 10^{-13} \text{ s.} \quad (\text{measured})$$

$$\Rightarrow \quad \begin{array}{c} \nu \\ \nearrow \\ \text{---} \bullet \text{---} \ell \\ \downarrow \text{wavy} \\ g_W \gamma^\mu (1 - \gamma^5) \end{array}$$

weak interaction in hadrons

$$B: \quad \begin{array}{c} p \\ \nearrow \\ n \text{---} \bullet \text{---} \\ \downarrow \text{wavy} \\ \frac{1}{q^2 - m_W^2} \\ \downarrow \text{wavy} \\ g_W \end{array} \quad \begin{array}{c} e^- \\ \leftarrow \\ \bar{\nu}_e \end{array}$$

using same vertex structure for lepton and hadrons

$$\mathcal{M} \sim g_W^2 \bar{\psi}_p \gamma^\mu (1 - \gamma^5) \psi_n \frac{1}{q^2 - m_W^2} \bar{\psi}_\ell \gamma_\mu (1 - \gamma^5) \psi_e$$

$$\text{initial state's} \quad n \quad \begin{array}{c} \Rightarrow \\ \longrightarrow \end{array} \quad \begin{array}{c} \Leftarrow \\ \longrightarrow \end{array}$$

$$\text{Final state's} \quad p \quad \begin{array}{c} \Rightarrow \\ \longrightarrow \end{array} \quad \begin{array}{c} \Leftarrow \\ \longrightarrow \end{array}$$

$$e^- \quad \begin{array}{c} \Rightarrow \\ \longrightarrow \end{array} \quad \begin{array}{c} \Leftarrow \\ \longrightarrow \end{array}$$

$$\bar{\nu}_e \quad \begin{array}{c} \Leftarrow \\ \longrightarrow \end{array}$$

$$\Gamma(n \rightarrow p e^- \bar{\nu}_e) = \frac{1}{\tau} \Rightarrow \tau^{\text{pred.}} = 1318 \text{ s.}$$

$$\text{Experimentally} \quad \tau_n^{\text{meas}} = 878 \text{ s} \neq \tau^{\text{pred.}}$$

$\Rightarrow$ hadrons behave differently.

$$\Rightarrow \text{hadron vertex} \quad \gamma^\mu (C_V + C_A \gamma^5)$$

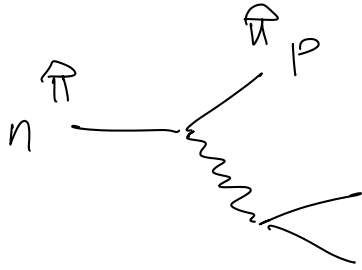
measure C_V, C_A for different hadrons.

$$\gamma^\mu (C_V + C_A \gamma^5)$$

$$\gamma^\mu (1 - \gamma^5)$$

C_V : vector coupling
 C_A : axial coupling.

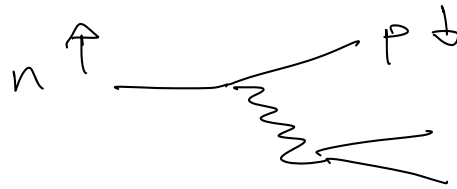
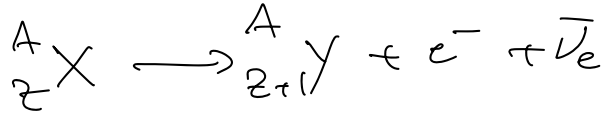
Different nuclear states



Same spin for p and n.

Fermi Transitions

Sensitive to C_V



Gamow-Teller Transitions

sensitive to C_V, C_A

$$\frac{d\Gamma}{dE_e}(x \rightarrow y e^- \bar{\nu}_e) = \underbrace{\left(\frac{g_w}{m_w}\right)^4}_{G_F^2} \times \underbrace{F(C_V, C_A)}_{\text{from matrix element}}$$

Experimental picture:

$$C_V \approx 1.000 \quad \text{same as leptons}$$

axial part depends on the hadrons.

$$\text{neutron: } C_A = -1.267 \quad (\text{obtained from polarized neutrons})$$

$$n \rightarrow p e^- \bar{\nu}_e \quad C_A/C_V = -1.267$$

$$\Lambda \rightarrow p \pi^-, n \pi^0 \quad = -0.718 \quad \Delta S = 1$$

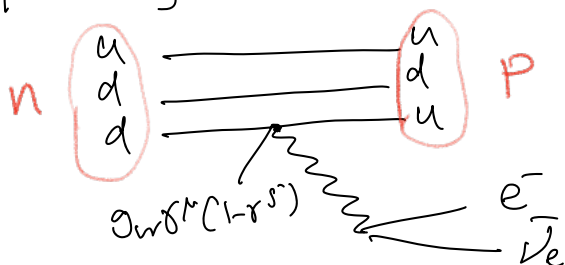
$$\Lambda \equiv uds$$

$$\Sigma^- \rightarrow n e^- \bar{\nu}_e \quad = +0.340$$

$$\Sigma^- \rightarrow \Lambda e^- \bar{\nu}_e \quad = -0.25$$

$\Rightarrow$ Consider weak interaction of quarks.

β decay in terms of quarks.



weak interaction of quarks.
+
strong int. corrections.

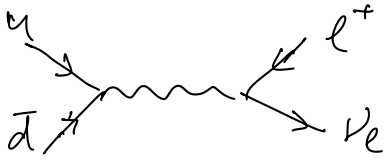
$\Rightarrow$ easier/simpler to use mesons (more than baryons)

$$\pi^+ = u\bar{d}$$

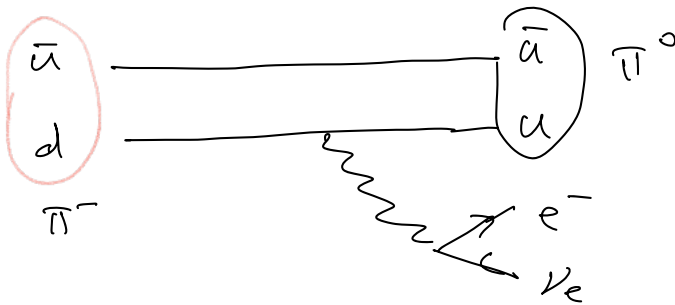
$$\pi^+ \rightarrow \mu^+ \nu_\mu$$

$$e^+ \nu_e$$

leptonic decays.



Semileptonic decay.



$$Q = m_{\pi^+} - m_{\pi^0} - m_e \gtrsim 0$$

$$m_{\pi^+} > m_{\pi^0}$$

very small phase space.

$$Q = 3.5 \text{ MeV}$$

π^+ DECAY MODES

π^- modes are charge conjugates of the modes below.

For decay limits to particles which are not established, see the section on Searches for Axions and Other Very Light Bosons.

Mode	Fraction (Γ_i/Γ)	Confidence level
$\Gamma_1 \mu^+ \nu_\mu$	[a] (99.98770 $\pm$ 0.00004) %	
$\Gamma_2 \mu^+ \nu_\mu \gamma$	[b] (2.00 $\pm$ 0.25) $\times 10^{-4}$	
$\Gamma_3 e^+ \nu_e$	[a] (1.230 $\pm$ 0.004) $\times 10^{-4}$	
$\Gamma_4 e^+ \nu_e \gamma$	[b] (7.39 $\pm$ 0.05) $\times 10^{-7}$	
$\Gamma_5 e^+ \nu_e \pi^0$	(1.036 $\pm$ 0.006) $\times 10^{-8}$	
$\Gamma_6 e^+ \nu_e e^+ e^-$	(3.2 $\pm$ 0.5) $\times 10^{-9}$	
$\Gamma_7 \mu^+ \nu_\mu \nu \bar{\nu}$	< 9	$\times 10^{-6}$ 90%
$\Gamma_8 e^+ \nu_e \nu \bar{\nu}$	< 1.6	$\times 10^{-7}$ 90%

Lepton Family number (LF) or Lepton number (L) violating modes

$\Gamma_9 \mu^+ \bar{\nu}_e$	L	[c] < 1.5	$\times 10^{-3}$	90%
$\Gamma_{10} \mu^+ \nu_e$	LF	[c] < 8.0	$\times 10^{-3}$	90%
$\Gamma_{11} \mu^- e^+ e^+ \nu$	LF	< 1.6	$\times 10^{-6}$	90%

Pion leptonic decay

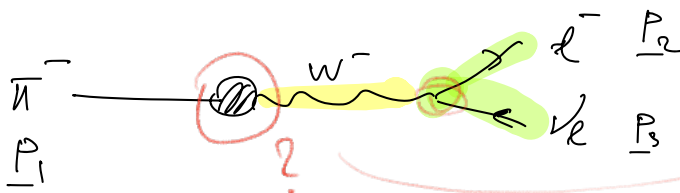
$$\pi^- \rightarrow e^- \bar{\nu}_e$$

$$Q = m_\pi - m_e = 135 \text{ MeV}$$

$$\mu^- \bar{\nu}_\mu$$

$$Q = m_\pi - m_\mu \approx 29 \text{ MeV}$$

1958 @ CERN by Fidecaro measured $\Gamma(\pi \rightarrow e) \neq 0$



$$\mathcal{M} \propto \frac{g_W^2}{q^2 - m_W^2} (\bar{u}(p_3) \gamma^\mu (1 - \gamma^5) v(p_2)) f_\pi p_{1\mu}$$

$\mathcal{M}$ is scalar

f_π should be a Lorentz vector.

$$\pi^- \text{ has } S=0. \Rightarrow F_\mu = f_\pi p_\mu^\pi = f_\pi p_{1\mu}$$

f_π : pion decay constant.

must be measured.

$$\langle |M|^2 \rangle = \sum_{\text{final state spins}} \sum_{\text{average init state spins}} |M|^2$$

initial state: $s=0 \rightarrow$

Final state: $e^- \leftarrow \Rightarrow \rightarrow \bar{\nu}_e$

$\bar{\nu}_\ell \rightarrow$

$$\langle |M|^2 \rangle = \left(\frac{g_w}{2m_w} \right)^4 f_\pi^2 m_e^2 (m_\pi^2 - m_e^2)$$

$\pi \rightarrow e \bar{\nu}_e$

$\Gamma_\pi \propto \langle |M|^2 \rangle \times \text{phase space.}$

$$E_\nu = |\vec{P}_\nu| = P_2$$

Ref frame of π at rest

$$\begin{array}{c} \pi \\ \leftarrow \quad \circ \quad \rightarrow \\ \vec{P}_e = -\vec{P}_\nu = \vec{P}_2 \end{array}$$

$\Rightarrow$ conservation of momentum

$$m_\pi = E_e + E_\nu = \sqrt{m_e^2 + P_2^2} + P_2$$

$$\Rightarrow P_2 = \frac{1}{2m_\pi} (m_\pi^2 - m_e^2)$$

$$\Gamma = \frac{1}{8\pi} \frac{|\vec{P}_2|}{m_\pi^2} \langle |M|^2 \rangle \quad \text{for } e^-, \bar{\nu}^-$$

$$\frac{\Gamma_e}{\Gamma_\mu} = \frac{\Gamma(\pi^- \rightarrow e^- \bar{\nu}_e)}{\Gamma(\pi^- \rightarrow \mu^- \bar{\nu}_\mu)} = \frac{m_e^2}{m_\mu^2} \frac{(m_\pi^2 - m_e^2)^2}{(m_\pi^2 - m_\mu^2)^2} = \frac{m_e^2}{m_\mu^2} \frac{m_\pi^2 (1 - \frac{m_e^2}{m_\pi^2})^2}{m_\mu^4 (1 - m_\mu^2/m_\pi^2)^2}$$

not elem $\times$ phase space

not elem.

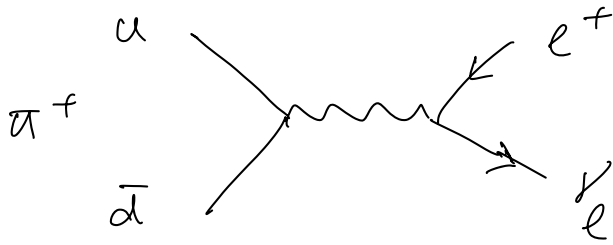
$$\left(\frac{m_e}{m_\mu} \right)^2 = \left(\frac{0.5 \text{ MeV}}{106 \text{ MeV}} \right)^2 \ll 1.$$

$$\frac{\Gamma_e}{\Gamma_\mu} = 1.283 \times 10^{-4} \quad \text{in agreement w/ experiments.}$$

$$e^- \xleftarrow{s=0} \circ \xrightarrow{\Rightarrow} \bar{\nu}_e$$

$$\mu^- \xleftarrow{\Leftarrow} \circ \xrightarrow{\Rightarrow} \bar{\nu}_\mu$$

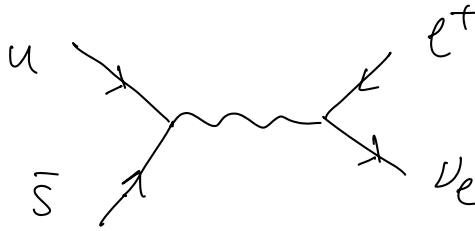
$$\beta_e = \frac{\Gamma_e}{\Gamma_\mu} = 0.38$$



$$\pi^+ \rightarrow e^+ \nu_e$$

$$m_\pi \approx 140 \text{ MeV}$$

$$m_\mu \approx 106 \text{ MeV}$$



$$K^+ \rightarrow e^+ \nu_e$$

$$m_K \approx 500 \text{ MeV}$$

π^+ DECAY MODES

π^- modes are charge conjugates of the modes below.

For decay limits to particles which are not established, see the section on Searches for Axions and Other Very Light Bosons.

Mode	Fraction (Γ_i/Γ)	Confidence level
Γ_1 $\mu^+ \nu_\mu$	[a] (99.98770 $\pm$ 0.00004) %	
Γ_2 $\mu^+ \nu_\mu \gamma$	[b] (2.00 $\pm$ 0.25) $\times 10^{-4}$	
Γ_3 $e^+ \nu_e$	[a] (1.230 $\pm$ 0.004) $\times 10^{-4}$	
Γ_4 $e^+ \nu_e \gamma$	[b] (7.39 $\pm$ 0.05) $\times 10^{-7}$	
Γ_5 $e^+ \nu_e \pi^0$	(1.036 $\pm$ 0.006) $\times 10^{-8}$	
Γ_6 $e^+ \nu_e e^+ e^-$	(3.2 $\pm$ 0.5) $\times 10^{-9}$	
Γ_7 $\mu^+ \nu_\mu \nu \bar{\nu}$	< 9 $\times 10^{-6}$	90%
Γ_8 $e^+ \nu_e \nu \bar{\nu}$	< 1.6 $\times 10^{-7}$	90%

Lepton Family number (LF) or Lepton number (L) violating modes

Γ_9 $\mu^+ \bar{\nu}_e$	L	[c] < 1.5 $\times 10^{-3}$	90%
Γ_{10} $\mu^+ \nu_e$	LF	[c] < 8.0 $\times 10^{-3}$	90%
Γ_{11} $\mu^- e^+ e^+ \nu$	LF	< 1.6 $\times 10^{-6}$	90%

K^+ DECAY MODES

Fraction (Γ_i/Γ)

Scale factor/ p
Confidence level (MeV/c)

Leptonic and semileptonic modes

$e^+ \nu_e$	(1.582 $\pm$ 0.007) $\times 10^{-5}$	247
$\mu^+ \nu_\mu$	(63.56 $\pm$ 0.11) %	S=1.2 236
$\pi^0 e^+ \nu_e$	(5.07 $\pm$ 0.04) %	S=2.1 228
Called K_{e3}^+		
$\pi^0 \mu^+ \nu_\mu$	(3.352 $\pm$ 0.033) %	S=1.9 215
Called $K_{\mu 3}^+$		
$\pi^0 \pi^0 e^+ \nu_e$	(2.55 $\pm$ 0.04) $\times 10^{-5}$	S=1.1 206
$\pi^+ \pi^- e^+ \nu_e$	(4.247 $\pm$ 0.024) $\times 10^{-5}$	203
$\pi^+ \pi^- \mu^+ \nu_\mu$	(1.4 $\pm$ 0.9) $\times 10^{-5}$	151
$\pi^0 \pi^0 \pi^0 e^+ \nu_e$	< 3.5 $\times 10^{-6}$	CL=90% 135

Hadronic modes

$\pi^+ \pi^0$	(20.67 $\pm$ 0.08) %	S=1.2 205
$\pi^+ \pi^0 \pi^0$	(1.760 $\pm$ 0.023) %	S=1.1 133
$\pi^+ \pi^+ \pi^-$	(5.583 $\pm$ 0.024) %	125

Cons: dkr $\Gamma(K^+ \rightarrow \mu^+ \nu_\mu)$

$$\pi^+ \text{ diagram: } u \text{ and } \bar{d} \text{ meet at vertex, exchange } W^+ \text{ boson, then split into } \mu^+ \text{ and } \nu_\mu.$$

$$\mu \sim f_\pi \frac{g_W^2}{m_W^2} m_e^2 (m_\pi^2 - m_e^2)$$

$$\Gamma_\pi \sim G_{F\pi}^2 f_\pi^2 \frac{m_e^2}{m_\pi^2} (m_\pi^2 - m_e^2)^2$$

$$K^+ \text{ diagram: } u \text{ and } \bar{s} \text{ meet at vertex, exchange } W^+ \text{ boson, then split into } \mu^+ \text{ and } \nu_\mu.$$

$$\mu \sim f_K \frac{g_W^2}{m_W^2} m_e^2 (m_K^2 - m_e^2)$$

$$\Gamma(K^+ \rightarrow \mu^+ \nu_\mu) \sim G_{FK}^2 f_K^2 \frac{m_e^2}{m_K^2} (m_K^2 - m_e^2)^2$$

$$\frac{\Gamma(K^+ \rightarrow \mu^+ \nu_\mu)}{\Gamma(\pi^+ \rightarrow \mu^+ \nu_\mu)} = \frac{G_{FK}^2}{G_{F\pi}^2} \left(\frac{f_K}{f_\pi}\right)^2 \left(\frac{m_\pi}{m_K}\right)^3 \frac{(m_K^2 - m_\mu^2)^2}{(m_\pi^2 - m_\mu^2)^2}$$

$$\Gamma(K^+ \rightarrow \mu^+ \nu_\mu) = \Gamma_{\text{tot}} \times \text{BF}(K^+ \rightarrow \mu^+ \nu_\mu)$$

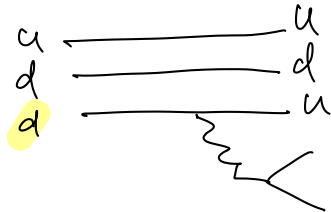
$$\#(K^+ \rightarrow \mu^+) / \# K_{\text{cons.}}$$

→ measured experimentally.

$$\left(\frac{G_F K}{G_F \pi} \right)^2 \approx 0.05$$

⇒ β decay of $\pi \neq K$ different rate.

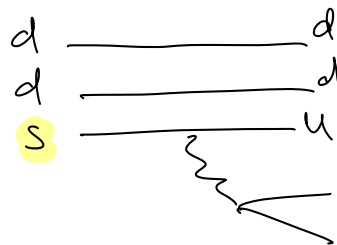
$$n \rightarrow p e^- \bar{\nu}_e$$



n DECAY MODES

Mode	Fraction (Γ_i/Γ)	Confidence level
Γ_1 $pe^- \bar{\nu}_e$	100 %	
Γ_2 $pe^- \bar{\nu}_e \gamma$	[a] $(9.2 \pm 0.7) \times 10^{-3}$	
Γ_3 hydrogen-atom $\bar{\nu}_e$	$< 2.7 \times 10^{-3}$	95%

$$\Sigma^- \rightarrow n e^- \bar{\nu}_e$$



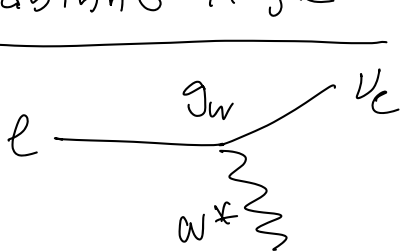
Σ^- DECAY MODES

Mode	Fraction (Γ_i/Γ)	Confidence level
Γ_1 $n\pi^-$	$(99.848 \pm 0.005) \%$	
Γ_2 $n\pi^- \gamma$	[a] $(4.6 \pm 0.6) \times 10^{-4}$	
Γ_3 $ne^- \bar{\nu}_e$	$(1.017 \pm 0.034) \times 10^{-3}$	
Γ_4 $n\mu^- \bar{\nu}_\mu$	$(4.5 \pm 0.4) \times 10^{-4}$	
Γ_5 $\Lambda e^- \bar{\nu}_e$	$(5.73 \pm 0.27) \times 10^{-5}$	
Γ_6 $\Sigma^+ X$	$< 1.2 \times 10^{-4}$	90%

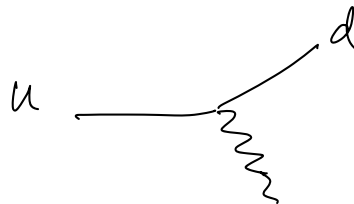
$$\Rightarrow \left(\frac{G_{Fs}}{G_{Fd}} \right)^2 \approx 0.05 \Rightarrow g_w^d \neq g_w^s ?$$

solution: Cabibbo angle

Cabibbo Angle



W^- couples to $\begin{pmatrix} e^- \\ \bar{\nu}_e \end{pmatrix}_L$



W^- couples to $\begin{pmatrix} u \\ d' \end{pmatrix}_L$

$$d' = \cos\theta \cdot d + \sin\theta \cdot s$$

θ : Cabibbo angle

mixing angle between d, s quarks

Nature produces u, d, s quarks.

weak interaction has a different basis than flavor in QCD

Flavor eigenstates $\neq$ weak eigenstates.

Nicola Cabibbo
CERN, Geneva, Switzerland
(Received 29 April 1963)

We present here an analysis of leptonic decays based on the unitary symmetry for strong interactions, in the version known as "eightfold way,"¹ and the V-A theory for weak interactions.^{2,3} Our basic assumptions on J_μ , the weak current of strong interacting particles, are as follows:

(1) J_μ transforms according to the eightfold representation of SU_3 . This means that we neglect currents with $\Delta S = -\Delta Q$, or $\Delta I = 3/2$, which should belong to other representations. This limits the scope of the analysis, and we are not

able to treat the complex of K^0 leptonic decays, or $\Sigma^+ \rightarrow n + e^+ + \nu$ in which $\Delta S = -\Delta Q$ currents play a role. For the other processes we make the hypothesis that the main contributions come from that part of J_μ which is in the eightfold representation.

(2) The vector part of J_μ is in the same octet as the electromagnetic current. The vector contribution can then be deduced from the electromagnetic properties of strong interacting particles. For $\Delta S = 0$, this assumption is equivalent to vector-

To determine θ , let us compare the rates for $K^+ \rightarrow \mu^+ + \nu$ and $\pi^+ \rightarrow \mu^+ + \nu$; we find

$$\Gamma(K^+ \rightarrow \mu\nu) / \Gamma(\pi^+ \rightarrow \mu\nu) = \tan^2 \theta M_K^2 (1 - M_\mu^2 / M_K^2)^2 / M_\pi^2 (1 - M_\mu^2 / M_\pi^2)^2. \quad (3)$$

From the experimental data, we then get^{5,6}

$$\theta = 0.257. \quad (4)$$

For an independent determination of θ , let us consider $K^+ \rightarrow \pi^0 + e^+ + \nu$. The matrix element for this process can be connected to that for $\pi^+ \rightarrow \pi^0 + e^+ + \nu$, known from the conserved vector-current hypothesis (2nd assumption). From the rate⁶ for $K^+ \rightarrow \pi^0 + e^+ + \nu$, we get

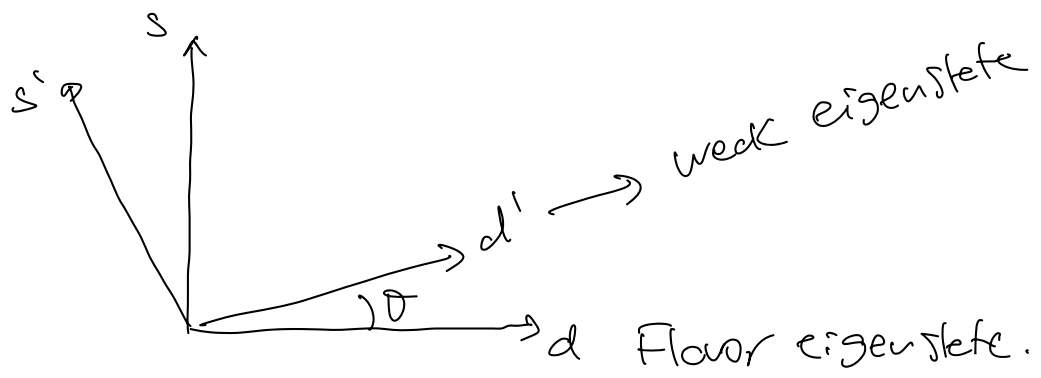
$$\theta = 0.26. \quad (5)$$

The two determinations coincide within experimental errors; in the following we use $\theta = 0.26$.

We go now to the leptonic decays of the baryons, of the type $A \rightarrow B + e + \nu$. The matrix element of

$$\frac{\Gamma(K^+ \rightarrow \mu^+ \nu)}{\Gamma(\pi^+ \rightarrow \mu^+ \nu)} = (\tan \theta)^2 \underbrace{F(m_\mu, m_u, m_K, f_\pi, f_K)}_{\text{pure algebraic factor.}}$$

$\hookrightarrow 0.05 \Rightarrow \theta = 0.25^\circ$



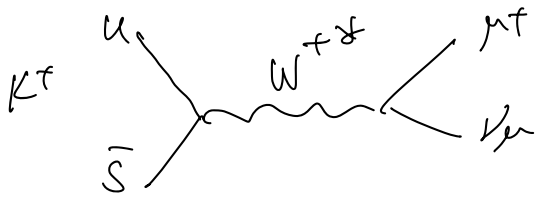
$$d' = d \cdot \cos\theta + S \cdot \sin\theta$$

$$S' = -\sin\theta \cdot d + \cos\theta \cdot S$$

$$\Gamma(\Delta S = 1) \propto \sin^2\theta \quad \text{Cabibbo-suppressed decays.}$$

$$\Gamma(\Delta S = 0) \propto \cos^2\theta \quad \text{Cabibbo-favored.}$$

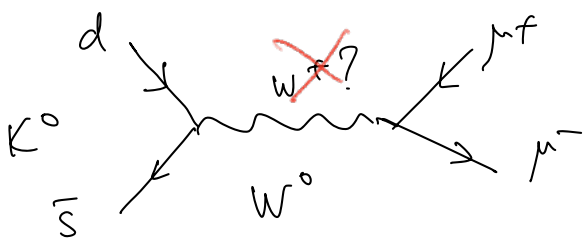
One other major problem with weak decays!



$$\Gamma(K^+ \rightarrow \pi^+ \nu_\mu) \simeq 64\%$$

$$K^+ \rightarrow \pi^+ \nu_\mu \quad Q = m_K - m_\mu \simeq 394 \text{ MeV.}$$

$$K^0 \rightarrow \pi^+ \pi^- \quad Q = m_K - 2m_\mu \simeq 288 \text{ MeV}$$



$$\Gamma(K^0 \rightarrow \pi^+ \pi^-) \simeq 10^{-9} \neq 0$$

neutral flavor changing weak current

Does W^0 exist?

$$\text{If } W^0 \text{ exists} \Rightarrow \Gamma(K^0 \rightarrow \pi^+ \pi^-) \simeq \Gamma(K^+ \rightarrow \pi^+ \nu_\mu)$$

$$\text{if } \Gamma(K^0 \rightarrow \pi^+ \pi^-) \ll \Gamma(K^+ \rightarrow \pi^+ \nu_\mu)$$

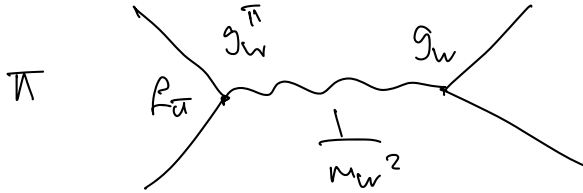
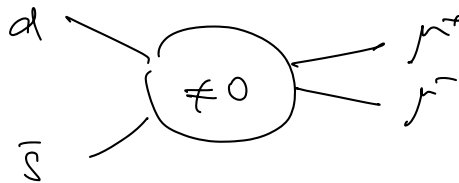
$$\Rightarrow m_{W^0} \gg m_{W^\pm}$$

$$\text{if } W^0 \text{ exists} \Rightarrow W^0 \rightarrow e^+ e^-, \nu_e \bar{\nu}_e$$



$K^+ \rightarrow \pi^+ \ell \bar{\ell}$
not seen.

$$K^0 \rightarrow \mu^+ \mu^- \neq 0$$



$$\mathcal{M} \sim g_w^{\pi} g_w \frac{1}{m_w^2}$$

$$\mathcal{M}^2 \sim g_w^{\pi^2} \frac{g_w^2}{m_w^2}$$

$$\underbrace{G_F^{\pi^2}}_{G_F^{\pi^2}}$$

$$\Gamma(\tau^- \rightarrow \mu^- \nu_{\mu} \nu_{\tau}) = \Gamma_{tot}$$

$$BF(\tau^- \rightarrow \mu^- \nu \bar{\nu}) = \frac{\Gamma(\tau^- \rightarrow \mu^- \nu \bar{\nu})}{\Gamma_{tot}} = \tau_{\tau} \cdot \Gamma(\tau^- \rightarrow \mu^- \nu \bar{\nu})$$

$$\frac{\# \tau^- \rightarrow \mu^-}{\# \text{ of } \tau} \quad \Gamma(\tau^- \rightarrow \mu^-) = \frac{BF(\tau^- \rightarrow \mu^-)}{\tau_{\tau}}$$

K⁺ DECAY MODES

Fraction (Γ_i/Γ)

Scale factor/
Confidence level (MeV/c)

Leptonic and semileptonic modes

$e^+ \nu_e$	(1.582 ± 0.007) × 10 ⁻⁵		247
$\mu^+ \nu_{\mu}$	(63.56 ± 0.11) %	S=1.2	236
$\pi^0 e^+ \nu_e$	(5.07 ± 0.04) %	S=2.1	228
Called K_{e3}^+ .			
$\pi^0 \mu^+ \nu_{\mu}$	(3.352 ± 0.033) %	S=1.9	215
Called $K_{\mu3}^+$.			
$\pi^0 \pi^0 e^+ \nu_e$	(2.55 ± 0.04) × 10 ⁻⁵	S=1.1	206
$\pi^+ \pi^- e^+ \nu_e$	(4.247 ± 0.024) × 10 ⁻⁵		203
$\pi^+ \pi^- \mu^+ \nu_{\mu}$	(1.4 ± 0.9) × 10 ⁻⁵		151
$\pi^0 \pi^0 \pi^0 e^+ \nu_e$	< 3.5 × 10 ⁻⁶	CL=90%	135

Hadronic modes

$\pi^+ \pi^0$	(20.67 ± 0.08) %	S=1.2	205
$\pi^+ \pi^0 \pi^0$	(1.760 ± 0.023) %	S=1.1	133
$\pi^+ \pi^+ \pi^-$	(5.583 ± 0.024) %		125